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# Chapter 1

# Measure Theory

## 1.1 Bernoulli sequences

Let $\mathcal{B}$ be the set of all Bernoulli sequences. $\mathcal{B}$ is uncountable.

**Proposition 1.1.** *If we delete a countable subset of $\mathcal{B}$, we can index what is left by the points on the real interval $I = ]0, 1]$. That is, there exists an injective function $f : I \to \mathcal{B}$.*

*Proof.* Each $\omega \in I$ can be written as

$$\begin{aligned} \omega &= \sum_{i=1}^{\infty} \frac{a_i}{2^i} \qquad a_i = 0, 1 \\ &= 0.a_1 a_2 \ldots \end{aligned}$$

Since $\omega$ may not have a unique binary representation, we will only consider non-terminating expansion for $\omega$. Then, by mapping 1 to $H$ and 0 to $T$ we get an injective function from $I$ to $\mathcal{B}$. To show this, suppose $\mathcal{B}_{\text{deg}}$ is the set of all Bernoulli sequences that after a certain point degenerates to all tails.

**Lemma 1.2.** *$\mathcal{B}_{\text{deg}}$ is countable.*

*Proof.* Let $\mathcal{B}^k_{\text{deg}}$ be all degenerate Bernoulli sequences where we have only tails after $k_{\text{th}}$ toss. Then, $\mathcal{B}^k_{\text{deg}}$ is finite and

$$\mathcal{B}_{\text{deg}} = \bigcup_{k=1}^{\infty} \mathcal{B}^k_{\text{deg}}$$

is the countable union of finite set and hence $\mathcal{B}_{\text{deg}}$ is countable. □

This concludes the proof. ■

**Definition (Borel Principle):** Suppose $E$ is a probabilistic event occuring in certain sequences. Let $\mathcal{B}_E$ denote the subset of $\mathcal{B}$ for which that event occurs. Let $I_E$ be the corresponding subset of $I$, then

$$\mathbb{P}(E) = \mu_L(I_E)$$

where $\mu_L$ is Lebesgue measure.

**Example 1.1.** Start with $X$ dollars and at each toss you win 1 dollars if heads shows up and ypu lose 1 dollars if tail shows up. What is the probability you lose all your original stake?

For $\omega \in I$ define the $k_{\text{th}}$ *Radamcher* function, $R_k$, by

$$\begin{aligned} R_k(\omega) &= 2a_k - 1 \\ &= \begin{cases} +1 & a_k = 1 \\ -1 & a_k = 0 \end{cases} \end{aligned}$$

Then, let $S_k(\omega)$ be the total amount won or loss at $k_{\text{th}}$ toss.

$$S_k(\omega) = \sum_{l=1}^{k} R_l(\omega)$$

Thus, the event that we lose our stake at $k_{\text{th}}$ is

$$I_{E_k} = \left\{ \omega \in I \,\middle|\, S_l(\omega) > -X \text{ for } l < k, S_k(\omega) = -X \right\}$$

hence

$$I = \bigcup_{l=1}^{\infty} I_{E_k}$$

is the event that we loss all our money eventually. We will however postpone calculating $\mu_L(I_E)$ as it is not finite union of intervals.

## 1.2 Weak law of large numbers

For some fixed $\epsilon$ let

$$I_N = \left\{ \omega \in I \,\middle|\, \left| \frac{s_N(\omega)}{N} - \frac{1}{2} \right| > \epsilon \right\}$$

where $s_N(\omega) = a_1 + a_2 + \cdots + a_N$. Then, $I_N$ represents the event that the number of heads and tails <u>are not</u> roughly equal.

**Theorem 1.3 (Weak law of large numbers).** *WLLN states*

$$\mu_L(I_N) \to 0 \text{ as } N \to \infty$$

*Proof.* Equivalently, for

$$A_N = \left\{ \omega \in I \,\middle|\, |S_N(\omega)| > 2N\epsilon \right\}$$

then

$$\mu_L(A_N) \to 0 \text{ as } N \to \infty$$

**Lemma 1.4 (Chebyshev's inequality).** *Let $f$ be a non-negative, piecewise constant function on $]0,1]$. Let $\alpha > 0$ be given. Then,*

$$\mu_L\left( \left\{ \omega \in I \,\middle|\, f(\omega) > \alpha \right\} \right) < \frac{1}{\alpha} \int_0^1 f \,\mathrm{d}x$$

*where $\int$ is the Riemann integral.*

*Proof.* Since $f$ is piecewise constant then there is a parition $0 = x_1 < \cdots < x_k = 1$ such that $f = c_i$ on $]x_i, X_{i+1}]$ for $i = 1, \ldots k-1$. Then,

$$\begin{aligned}\int_0^1 f \,\mathrm{d}x &= \sum_{i=1}^{k-1} c_i(x_{i+1} - x_i) \\ &\geq \sum_{c_i > \alpha} c_i(x_{i+1} - x_i) \\ &> \alpha \sum_{c_i > \alpha} x_{i+1} - x_i = \alpha \mu_L\Big(\Big\{\omega \in I \,\Big|\, f(\omega) > \alpha\Big\}\Big)\end{aligned}$$

which proves the Chebyshev's inequality. □

Then, we have

$$A_N = \Big\{\omega \in I \,\Big|\, |S_N(\omega)| > 2N\epsilon\Big\} = \Big\{\omega \in I \,\Big|\, (S_N(\omega))^2 > 4N^2\epsilon^2\Big\}$$

and hence

$$\begin{aligned}\mu_L(A_N) &< \frac{1}{4N^2\epsilon^2} \int_0^1 (S_N(\omega))^2 \,\mathrm{d}\omega \\ &= \frac{1}{4N^2\epsilon^2}\left[\sum_i \int_0^1 (R_i(\omega))^2 \,\mathrm{d}\omega + \sum_{i \neq j} \int_0^1 R_i(\omega) R_j(\omega) \,\mathrm{d}\omega\right] \\ &= \frac{1}{4N^2\epsilon^2} N = \frac{1}{4N\epsilon^2}\end{aligned}$$

Therefore, as $N$ approaches infinity, $\mu(A_N)$ approaches zero. ■

Now we want to show that for a "typical" Bernoulli sequence

$$\frac{1}{2} - \frac{s_N(\omega)}{N} \to 0 \text{ as } N \to \infty \tag{1.1}$$

and by "typical" we mean Equation (1.1) fails on a set of zero proability or the equivalent event has measure zero.

**Definition:** A set $A \subset \mathbb{R}$ has Lebesgue measure zero if for every $\epsilon > 0$, there exists a countable covering $\{A_i\}$ of $A$ by intervals such that

$$\sum_{i=1}^{\infty} \mu_L(A_i) < \epsilon$$

- Subset of a measure zero are measure zero.
- A signle point has a measure zero.
- countable union of measure zeros is a measure zero.

Let $N = \Big\{\omega \in I \,\Big|\, \frac{s_N(\omega)}{N} \to \frac{1}{2}\Big\}$. $N$ is called the set of **normal numbers**. Let $N^c$ be the complement of $N$.

**Theorem 1.5 (Strong law of large numbers).** *SLLN states that $N^c$ has measure zero.*

*Proof.* Let $A_n = \{\omega \in I \mid |S_n(\omega)| > \epsilon n\}$ then

$$A_n = \left\{\omega \in I \,\middle|\, (S_n(\omega))^4 > \epsilon^4 n^4\right\}$$

By Chebyshev's inequality

$$\begin{aligned}\mu_L(A_n) &< \frac{1}{n^4\epsilon^4}\int_0^1 (S_n(\omega))^4 \,\mathrm{d}\omega \\ &= \frac{1}{n^4\epsilon^4}(n + 3n(n-1)) \leq \frac{3}{\epsilon^4 n^2}\end{aligned}$$

**Lemma 1.6.** *Given $\delta > 0$ there exists a sequence $\epsilon_1, \epsilon_2, \ldots$ such that $\epsilon \to 0$ and*

$$\sum_{n=1}^{\infty} \frac{3}{n^2\epsilon_n^4} < \delta$$

*Proof.* Choose $\epsilon_n^4 = cn^{-1/2}$ for some constant $c$.

$$\sum_{n=1}^{\infty} \frac{3}{n^2\epsilon_n^4} = \frac{3}{c}\sum_{n=1}^{\infty} n^{-\frac{3}{2}} = \frac{3}{c}L < \delta$$

if $c > \dfrac{3L}{\delta}$. □

Finally, Let $B_n = \{\omega \in I \mid |S_n(\omega)| > \epsilon_n n\}$. Then, by our first computation

$$\mu_L(B_n) < \frac{3}{\epsilon_n^4 n^2}$$

and by the lemma

$$\sum \mu_L(B_n) < \frac{3}{\epsilon_n^4 n^2} < \delta$$

It remains to show that $N^c \subset \cup_{n=1}^{\infty} B_n$ which is obvious. As every $\omega \in N^c$ is some $B_n$. Therefore, $N^c$ has measure zero. ■

## 1.3 Measure theory

### 1.3.1 Measure

**Definition:** A **ring** of sets in $X$ is a non-empty collection $\mathscr{R}$ of subsets of $X$ satisfying following two properties

1. $A, B \in \mathscr{R} \implies A \cup B \in \mathscr{R}$.
2. $A, B \in \mathscr{R} \implies A - B \in \mathscr{R}$.

**Lemma 1.7.** *If $A, B \in \mathscr{R}$, then $A \cap B \in \mathscr{R}$. Moreover, $R$ is a ring if and only if it is closed under intersection and symmetric difference.*

*Proof.*

$$A \cap B = [(A \cup B) - (A - B)] - (B - A)$$

■

**Example 1.2.** $\mathcal{P}(X)$ is a ring.

**Definition:** A **semi-ring** is a non-empty collection of $\mathscr{S}$ such that

1. $\emptyset \in \mathscr{S}$.
2. $A, B \in \mathscr{S} \implies A \cap B \in \mathscr{S}$.
3. $A, B \in \mathscr{S} \implies A - B = \cup_{i=1}^{k} C_i$ where $C_i \in \mathscr{S}$ are disjoint.

The ring generated by a semi-ring $\mathscr{S}$ is denoted by $\mathcal{R}(\mathscr{S})$.

**Theorem 1.8.** *Suppose $\mathscr{S}$ is a semi-ring. Then, $A \in \mathcal{R}(\mathscr{S})$ if and only if $A = \cup_{i=1}^{k} C_i$ for disjoint $C_i \in \mathscr{S}$.*

**Example 1.3.** Let $\mathscr{R}_{\text{Leb}} = \{A \mid A = \bigcup_{i=1}^{n} A_i\}$ where $A_i$ are disjoint $k$-cells in $\mathbb{R}^k$. $\mathscr{R}_{\text{Leb}}$ is a ring. Why?

**Definition:** Let $\mu : \mathscr{R} \to \overline{\mathbb{R}}_0^+$ where $\mathscr{R}$ is a ring of some set $X$, then $\mu$ is **additive** if $\mu(A \cup B) = \mu(A) + \mu(B)$ whenever $A, B \in \mathscr{R}$ are disjoint.

**Proposition 1.9.** *Let $\mathscr{R}$ be a ring of $X$ and $\mu : \mathscr{R} \to \overline{\mathbb{R}}_0^+$ is additive, then*

1. $\mu(\emptyset) = 0$.
2. *(Monotonicity) If $A, B \in \mathscr{R}$ with $A \subset B \implies \mu(A) \geq \mu(B)$.*
3. *(Finite Addtivity) For disjoint $A_1, \ldots, A_n \in \mathscr{R}$, $\mu(\cup_{i=1}^{n} A_i) = \sum_{i=1}^{n} \mu(A_i)$.*
4. *(Lattice property) $A, B \in \mathscr{R} \implies \mu(A \cup B) + \mu(A \cap B) = \mu(A) + \mu(B)$.*
5. *(Finite subaddtivity) If $A_1, \ldots, A_n \in \mathscr{R}$, then*

$$\mu\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} \mu(A_i)$$

**Definition:** $\mu$ is **countably additive** on $\mathscr{R}$ if given any countable collection $\{A_i\} \subset \mathscr{R}$ with $A_i$ mutually disjoint and such that $A = \cup A_i$ is also in $\mathscr{R}$

$$\mu(A) = \sum_{i=1}^{\infty} \mu(A_i)$$

A countably additive, non-negative set function $\mu$ on ring $\mathscr{R}$ in $X$ is called a **measure**.

**Theorem 1.10.** *If $X = \mathbb{R}^n$, $\mathscr{R} = \mathscr{R}_{\text{Leb}}$ and $\mu$ for $n$-cells is defined as*

$$\mu(A) = \prod_{i=1}^{n} (b_i - a_i)$$

*where $A = \{x \in \mathbb{R}^n \mid x_i \in (a_i, b_i)\}$– $(a, b)$ denotes any of four possibilities, $]a, b[, ]a, b], [a, b[$ and $[a, b]$. Then, $\mu$ is a measure.*

To prove this theorem, we consider the following lemma

**Lemma 1.11.** *Let $A \in \mathscr{R}_{\text{Leb}}$ and let $\epsilon > 0$. There exists $F, G \in \mathscr{R}_{\text{Leb}}$ such that $F$ is closed and $G$ is open, $F \subset A \subset G$ and*

$$\mu(F) \geq \mu(A) - \epsilon$$
$$\mu(G) \leq \mu(A) + \epsilon$$

*Proof.*

*Proof.*

**Definition:** Let $\{A_n\}$ be a sequence of sets in $X$. Then,

$$\limsup A_n = \bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} A_n \qquad \liminf A_n = \bigcup_{k=1}^{\infty} \bigcap_{n=k}^{\infty} A_n$$

$A_n$ is said to converge to $A$ if $\limsup A_n = \liminf A_n = A$. $A$ is said to be **limit set** of $\{A_n\}$. The sequence $\{A_n\}$ is increasing if $A_n \subset A_{n+1}$ for all $n$ and it is decreasing if $A_n \supset A_{n+1}$ for all $n$.

It can be readily seen that if $\{A_n\}$ is increasing/decreasing, then it is convergent to $\cup A_n / \cap A_n$.

**Definition:** Let $\mathscr{R}$ be a ring of subsets in $X$ and $\mu : \mathscr{R} \to \overline{\mathbb{R}}_0^+$ is a set function. For any $E \in \mathscr{R}$, $\mu$ is said to be **continuous from below** if for all increasing sequences $E_n$, $\mu(E_n) \to \mu(E)$. Similarly, $\mu$ is said to be **continuous from above** if for all decreasing sequences $E_n$ such that $\mu(E_i) < \infty$ for at least one $i$, $\mu(E_n) \to \mu(E)$. $\mu$ is continuous at $E$ if it is both continuous from below and above.

**Theorem 1.12.** *A measure $\mu$ is continuous at every $E \in \mathscr{R}$.*

*Proof.*

**Proposition 1.13.** *Suppose $\mathscr{R}$ is ring of subsets of $X$ and $\mu$ is a finite additive function.*

- *If $\mu : \mathscr{R} \to \overline{\mathbb{R}}_0^+$ is continuous from below at every $E \in \mathscr{R}$, then $\mu$ is a measure.*
- *If $\mu : \mathscr{R} \to \mathbb{R}_0^+$ is continuous from above at $\emptyset$, then $\mu$ is a measure.*

### 1.3.2 Caratheodory extension

**Definition:** Let $A$ be a subset of $X$. A number $l \geq 0$ is called an **approximate outer measure** of $A$ if there exists a covering of $A$ by countable collection of sets $\{A_i\}$ with each $A_i \in \mathscr{R}$ such that

$$\sum_{i=1}^{\infty} \mu(A_i) \geq l$$

**Remark 1.** $l$ is allowed to be $+\infty$.

**Definition:** Let $A$ be a subset of $X$. The **outer measure** of $A$ is the greates lower bound of the set $\{l \mid l \text{ isanapproximateoutermeasure}\}$.

$$\mu^*(A) = \inf\left\{ l \,\middle|\, A \subset \bigcup_{i=1}^{\infty} A_i, \sum_{i=1}^{\infty} \mu(A_i) \leq l \right\}$$

If the set is empty, then $\mu^*(A) = +\infty$.

**Remark 2.** $\mu^* : \mathcal{P}(X) \to \overline{\mathbb{R}}_0^+$ is not a measure. However, $\mu^*$ is a measure on larger ring of subsets of $X$.

**Proposition 1.14.**

1. *If $A \in \mathscr{R}$, then $\mu^*(A) = \mu(A)$.*
2. *If $A \subset B$, then $\mu^*(A) \leq \mu^*(B)$.*
3. *$\mu^*$ is countably subadditive.*

$$\mu^*\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu^*(A_i)$$

*Proof.*

**Definition:** A set $A \subset X$ is **measurable** with respect to $\mu$ if for all $E \subset X$,

$$\mu(E) = \mu(E \cap A) + \mu(E \cap A^c)$$

The set of all measurable sets with respect to $\mu$ is denoted by $\mathcal{M}(\mu)$.

**Proposition 1.15.** *Let $\mu$ be a measure defined on the ring $\mathscr{R}$. Then, $\mathcal{M}(\mu^*)$ is a ring and $\mathscr{R} \subset \mathcal{M}(\mu^*)$.*

*Proof.* $\mathcal{M}(\mu^*)$ is closed under complementing and intersection. ■

**Definition:** Let $\mathscr{S}$ be a collection of subsets of a set $X$. $\mathscr{S}$ is called a **$\sigma$-ring** if

1. $\mathscr{S}$ is a ring.
2. $\mathscr{S}$ is closed under countable union. That is, given $\{A_i\} \subset \mathscr{S}$, $\bigcup A_i \in \mathscr{S}$.

**Theorem 1.16.** *$\mathcal{M}(\mu^*)$ is a $\sigma$-ring and the restriction $\mu^*$ on $\mathcal{M}(\mu^*)$ is a measure.*

**Remark 3.** The extension of $\mu$ to $\mu^*$ is not necessarily unique. However, by placing certain requirements on $X$ we can deduce uniqueness.

**Definition:** Let $\mathscr{R}$ be a ring of subsets in $X$ and $\mu : \mathscr{R} \to \overline{\mathbb{R}}_0^+$ be a measure. $\mu$ is **finite** if for each $A \in \mathscr{R}$, $\mu(A) < \infty$. $\mu$ is **$\sigma$-finite** if for each $E \in \sigma(R)$, there exists a sequence of subsets $\{E_n\} \subset \mathscr{R}$ such that $E \subset \cup E_n$ and $\mu(E_n) < \infty$ for all $n$.

**Definition:** A collection of subsets $\mathscr{C}$ is a **monotone class** if the limit set of every increasing and decreasing sequence of $\mathscr{C}$ is in $\mathscr{C}$. The smallest monotone class of a collection $\mathscr{D}$ is denoted by $\mathcal{C}(\mathscr{D})$.

**Theorem 1.17.** *If $\mathscr{R}$ is a ring and $\mathscr{C}$ is a monotone class containing $\mathscr{R}$, then $\sigma(\mathscr{R}) \subset \mathscr{C}$. In fact $\sigma(\mathscr{R}) = \mathcal{C}(\mathscr{R})$.*

**Corollary 1.18.** *Let $\mathscr{R}$ be a ring of subsets of $X$ and $\mu, \nu : \sigma(\mathscr{R}) \to \overline{\mathbb{R}}_0^+$ are two measures. If $\mu$ and $\nu$ are finite and equal on $\mathscr{R}$, then they are equal on $\sigma(\mathscr{R})$.*

**Definition:** Let $\mathscr{D}$ be a collection of subset of $X$ and $A \subset X$. Then

$$\mathscr{D}|_A = \{A \cap E \mid E \in \mathscr{D}\}$$

**Proposition 1.19.** *Let $\mathscr{R}$ be a ring. Then, $\sigma(\mathscr{R}|_A) = \sigma(\mathscr{R})|_A$.*

**Theorem 1.20.** *Suppose $\mathscr{R}$ is a ring of subsets of $X$ and $\mu : \mathscr{R} \to \overline{\mathbb{R}}_0^+$ is a $\sigma$-finite measure. The restriction of $\mu^*$ to $\sigma(\mathscr{R})$ is the only extension of $\mu$ to $\mathscr{R}$.*

### 1.3.3 Metric extension

Let $\mu : \mathscr{R} \to \mathbb{R}_0^+$ be a measure on the ring $\mathscr{R}$. For $A, B \in \mathcal{P}(X)$, we define a *psuedo distance* function on $\mathcal{P}(X)$

$$d(A, B) = \mu^*(A \triangle B)$$

Note that, $d(A, B)$ may be $+\infty$ and $d(A, B) = 0$ does not imply that $A = B$. To go around this constraint, we consider the equivalence relation $\sim$ with $A \sim B$ when $d(A, B) = 0$. Then, $d$ is metric on the equivalence classes $\mathcal{P}(X)/\sim$.

**Proposition 1.21.** *Suppose $A, B, C \in \mathcal{P}(X)$, then*

1. $d(A, B) = d(B, A)$.
2. $d(A, A) = 0$.
3. $d(A, B) \leq d(A, C) + d(C, A)$

*Proof.*

Although, $d$ is not quite a metric, we can still definte the notion of convergence; $A_i \to A$ if $d(A, A_i) \to 0$.

**Proposition 1.22.** *The Boolean operation in $\mathcal{P}(X)$ are continuous with respect to $d$. That is, if $A_n \to A$ and $B_n \to B$*

$$A_n \cup B_n \to A \cup B$$
$$A_n \cap B_n \to A \cap B$$
$$A_n^c \to A^c$$

**Proposition 1.23.** *$\mu^*$ is continuous in the following sense that for $A, B \in \mathcal{P}(X)$ where $\mu^*(A)$ or $\mu^*(B)$ is finite*

$$|\mu^*(A) - \mu^*(B)| \leq d(A, B)$$

**Definition:** Let $\mathscr{M}_F$ be the closure of $\mathscr{R}$ in $\mathcal{P}(X)$. That is, $A \in \mathscr{M}_F$ whenever there exists a sequence $\{A_i\} \subset \mathscr{R}$ such that $d(A_i, A) \to 0$ as $i \to \infty$.

**Theorem 1.24.**

1. *$\mathscr{M}_F$ is a ring.*
2. *For $A \in \mathscr{M}_F$, $\mu^*(A) < +\infty$.*
3. *$\mu^*$ is a measure on $\mathscr{M}_F$.*

**Definition:** $A$ is **measurable set**, $A \in \mathscr{M}$, if there exists a sequence $\{A_i\} \subset \mathscr{M}_F$ such that $A = \bigcup A_i$.

**Theorem 1.25.** *If $A \in \mathscr{M}$, then $A \in \mathscr{M}_F \iff \mu^*(A) < +\infty$.*

**Theorem 1.26.** *$\mathscr{M}$ is a $\sigma$-ring.*

**Theorem 1.27.** *If $A_1, A_2, \ldots$ is a countable collection of disjoint sets in $\mathscr{M}$ then*

$$\mu^*\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu^*(A_i)$$

*That is, $\mu^*$ is a measure on $\mathscr{M}$.*

We now investigate the relation between the measurablity in Caratheodory sense and metric sense. Note that, for measurablity in metric sese we assumed that $\mu$ is a finite measure. Therefore, we assume that $\mu^*$ is $\sigma$-finite with respect to $\mathscr{M}_F$.

**Theorem 1.28.** *Let $\mathscr{R}$ be a ring of subset of $X$ and $\mu : \mathscr{R} \to \mathbb{R}_0^+$ be a measure. If $A \in \mathscr{M}_F$, then for every $E \subset X$*

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

**Theorem 1.29.** *Let $\mathscr{R}$ be a ring of subset of $X$ and $\mu : \mathscr{R} \to \mathbb{R}_0^+$ be a measure. If $\mu^*(A) < \infty$ and for every $E \subset X$*

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

*then, $A \in \mathscr{M}_F$.*

Therefore, from the last two theorems we conclude that if the measure space of Caratheodory extension $(X, \mathscr{M}(\mu^*), \mu^*)$ is $\sigma$-finite, then both methods of extension result in the same extension.

### 1.3.4 Completion of measure spaces

**Lemma 1.30.** *Suppose $\mathscr{R}$ is a ring of subsets of $X$ and $\mu : \mathscr{R} \to \overline{\mathbb{R}}_0^+$ is measure. Furthermore, let $\mu^*$ be the outer measure of $\mu$. If $\mu^*(A) = 0$, then $A \in \mathcal{M}(\mu^*)$. ESpecially, for every subset $B \subset A$, $\mu^*(B) = 0$ and $B \in \mathcal{M}(\mu^*)$.*

**Definition:** A measure space $(X, \mathscr{F}, \mu)$ is **complete** if every subset of a null set, is in $\mathscr{F}$ and is measure zero.

**Theorem 1.31.** *Every measure space $(X, \mathscr{F}, \mu)$ can be uniquely extended to a complete measure space.*

Let $(X, \overline{\mathscr{F}}, \overline{\mu})$ be the extended complete measure described above. We shall investigate how $(X, \overline{\mathscr{F}}, \overline{\mu})$ is related to Caratheodory extension. Firstly, consider the following covering lemma.

**Lemma 1.32.** *Suppose $(X, \mathscr{F}, \mu)$ is a measure space. For every $E \subset X$, there exists a $A \in \mathscr{F}$ such that $E \subset C$ and $\mu^*(E) = \mu(C)$.*

**Theorem 1.33.** *Suppose $(X, \mathscr{F}, \mu)$ is a $\sigma$-finite measure space. If $(X, \overline{\mathscr{F}}, \overline{\mu})$ is the completion and $(X, \mathcal{M}(\mu^*), \mu^*)$ is the Caratheodory extension of $(X, \mathscr{F}, \mu)$, then $\overline{\mu} = \mu^*$ and $\overline{\mathscr{F}} = \mathcal{M}(\mu^*)$.*

Let $\mu$ be a measure on the ring $\mathscr{R}$ and $\mathscr{F} = \sigma(\mathscr{R})$ and let $\nu$ be the restriction of $\mu^*$ to $\mathscr{F}$.

**Theorem 1.34.** *For every $A \subset X$, $\mu^*(A) = \nu^*(A)$.*

Therefore, if $(X, \mathscr{F}, \nu)$ is $\sigma$-finite, then its completion is the same as $(X, \mathcal{M}(\mu^*), \mu^*)$

### 1.3.5 Lebesgue measure

**Example 1.4.** In the case of Lebesgue measure $\mu_L$ on $\mathscr{R}_{\text{Leb}}$, since it is a $\sigma$-finite measure, then its metric and Caratheodory extensions are equal. The restriction of $\mu_L^*$ to $\mathcal{M}(\mu_L^*)$, is called the **Lebesgue measure** and it is denoted by $\lambda_1 = \lambda$. The $\sigma$-field $\mathcal{M}(\mu_L^*)$ is called the **Lebesgue measurable sets** and it is denoted by $\Lambda^1 = \Lambda$.

**Proposition 1.35.** *Every open and closed subset of $\mathbb{R}^n$ is in $\mathscr{M}$.*

**Corollary 1.36.** *All countable unions and intersection of closed and open sets are measurable.*

**Definition:** The Borel sets, $\mathscr{B}$, is the $\sigma$-field generated by $\mathscr{R}_{\text{Leb}}$.

**Proposition 1.37.** *$\mathscr{B}$ contains all intervals and open sets. Moreover, it is the smallest $\sigma$-ring containing the open sets.*

**Theorem 1.38.** *If $A \in \Lambda$, there exists a Borel set $B \subset A$ such that $\lambda(A - B) = 0$. That is, $A$ can be written as $A = (A - B) \cup B$ where $B$ is Borel set and $\lambda(A - B) = 0$.*

**Theorem 1.39.** *For each $A \subset \mathbb{R}$ we have*

$$\lambda^*(A) = \inf\{\lambda(U) \,|\, A \subset U, U \text{ is open}\}$$

**Corollary 1.40.** *If $A \in \Lambda$ and if $\epsilon > 0$ is given, then there exists a Borel set such that $G \supset A$ and $\lambda(G - A) < \epsilon$.*

**Corollary 1.41.** *If $A \in \Lambda$, there exsists a Borel set $F \subset A$ with $\lambda(A - F) < \epsilon$.*

**Corollary 1.42.** *If $\mu$ is a measure on $\Lambda$ that for each Borel set $B$, $\mu(B) = \lambda(B)$, then $\mu = \lambda$.*

**Theorem 1.43.** *If $E$ is a Lebesgue measurable set, then*

$$\lambda(E) = \sup\{\lambda(K) \,|\, K \subset E, K \text{ is compact}\}$$

**Theorem 1.44.** *For each subset $A \subset \mathbb{R}$ and $c \in \mathbb{R}$, $\lambda^*(A + c) = \lambda^*(A)$ and $\lambda^*(cA) = |c|\lambda^*(A)$. Moreover, if $A$ is Lebesgue measurable, then $A + c$ and $cA$ are Lebesgue measurable as well.*

**Theorem 1.45.** *There exists a non-Lebesgue measurable set in $\mathbb{R}$.*

### 1.3.6 Finite signed measures

**Definition:** Suppose $(X, \mathscr{F})$ is measurable space. The set function $\nu : \mathscr{F} \to \mathbb{R}$ is a **finite signed measure** if it is countably additive. That is, for every sequence of disjoint subsets $\{A_n\}$, $\sum \nu(A_n)$ is convergent and

$$\nu\left(\bigcup_{i=1}^{\infty} A_n\right) = \sum_{i=1}^{n} \nu(A_n)$$

Since the order of right hand side summation does not matter, then the series is absolutely convergent.

**Proposition 1.46.**

1. $\nu(\emptyset) = 0$.
2. *$\nu$ is a finitely additive.*
3. *If $A, B \in \mathscr{F}$ and $A \subset B$, then $\nu(B - A) = \nu(B) - \nu(A)$.*
4. *$\nu$ is continuous from below at every $E \in \mathscr{F}$.*

**Definition:** Let $\nu$ be a finite signed measure on $\mathscr{F}$. For each $A \in \mathscr{F}$, the signed finite measure $\nu_A$ is defined as

$$\nu_A(E) = \nu(A \cap E)$$

**Proposition 1.47.** *$\nu_A$ is a measure if and only if for every subset $F \subset A$ that $F \in \mathscr{F}$, $\nu(F) \geq 0$.*

**Proposition 1.48.**

1. *$\nu_\emptyset(E) = 0$ for all $E \in \mathscr{F}$.*
2. *If $A, B \in \mathscr{F}$ and $A \cap B = \emptyset$, then $\nu_{A \cup B} = \nu_A + \nu_B$.*
3. *If $A, B \in \mathscr{F}$ and $A \subset B$, $\nu_{B-A} = \nu_B - \nu_A$.*
4. *If $A, B \in \mathscr{F}$, $\nu_{A \cap B} = (\nu_A)_B$.*
5. *If $A, B \in \mathscr{F}$, $\nu_{A \cup B} + \nu_{A \cap B} = \nu_A + \nu_B$.*

**Theorem 1.49 (Hann-Jordan decomposition).** *If $\nu$ is a signed finite measure on a $\sigma$-field $\mathscr{F}$, then there exists $A \in \mathscr{F}$ such that $\nu_A \geq 0$ and $\nu_{A^c} \leq 0$ hence*

$$\nu = \nu_A - (-\nu_{A^c})$$

*That is, $\nu$ is the difference of two finite measures.*

*Proof.* Let $\mathscr{N} = \{B \in \mathscr{F} \,|\, \nu_B \leq 0\}$. Then, $\mathscr{N}$ is closed under finite and countable union. Moreover if $B \in \mathscr{N}$ and $E \in \mathscr{F}$, then $B \cap E \in \mathscr{N}$. Consider the following lemma

**Lemma 1.50.** *The set $\{\nu(B) \,|\, B \in \mathscr{N}\}$ has an smallest element.*

then do some more work. ■

## 1.4 Measure theoretic modeling

**Definition:** Let $X$ be a set and $\mathscr{F}$ a ring of subsets of $X$.

1. $\mathscr{F}$ is a field if $X \in \mathscr{F}$.
2. $\mathscr{F}$ is a $\sigma$-field if $X \in \mathscr{F}$ and $\mathscr{F}$ is a $\sigma$-ring.

**Definition:** Let $X$ be a set and $\mathscr{F}$ be a field of subsets of $X$. Suppose $\mu$ is a measure defined on $\mathscr{F}$. Then, $\mu$ is a probability measure if $\mu(X) = 1$. In this case, the triplet $(X, \mathscr{F}, \mu)$ is a called probability space.

Let $X$ be a sample of space of a probabilistic process. A measure theoretic model of the proccess is a $\sigma$-field $\mathscr{F}$ of subsets of $X$ and probability measure $\mu$ defined on $\mathscr{F}$. So that, for any "plausible" event $E$ in $X$, we have $B_E \in \mathscr{F}$ and $\mathbb{P}(E) = \mu(B_E)$ where $B_E$ is the set of points in $X$ for which in $E$ occurs.

**Definition:** Given set $B_1, B_2, \ldots$ in $\mathscr{F}$, then

$$\{B_i;\ \text{i.o.}\} = \limsup B_n = \bigcap_{k=1}^{\infty} \bigcup_{n \geq k} B_n$$

**Theorem 1.51 (First Borel-Cantelli lemma).** *Given a sequence $B_1, B_2, \ldots$ in $\mathscr{F}$ define $B = \limsup B_n$. Then, $\sum_{i=1}^{\infty} \mu(B_i) < \infty$ implies $\mu(B) = 0$.*

**Definition:** Let $X$ be a sample space with $\sigma$-field $\mathscr{F}$ and probability measure $\mu$. Two sets $A_1, A_2 \in \mathscr{F}$ are **independent** if

$$\mu(A_1 \cap A_2) = \mu(A_1)\mu(A_2)$$

More generally, $A_1, \ldots, A_n$ are independent, if for any subset $I$ of $\mathbb{N}_n$

$$\mu\left(\bigcap_{i \in I} A_i\right) = \mathbb{P}(i \in I)\mu(A_i)$$

Furthermore, a countable collection of sets is independent if every finite subcollection is independent.

**Theorem 1.52 (Second Borel-Cantelli lemma).** *Assume $(X, \mathscr{F}, \mu)$ is a probability space and let $A_1, A_2, \ldots$ be an independent collection of sets from $\mathscr{F}$. Suppose that $\sum_{i=1}^{\infty} \mu(A_i)$ is not finite, then $\mu(\limsup A_n) = 1$.*

**Lemma 1.53.** *Let $A_1, A_2, \ldots$ be an independent collection of sets in $\mathscr{F}$. Then, $A_1^c, A_2^c \ldots$ is an independent collection of set in $\mathscr{F}$.*

# Chapter 2

# Integeration

**Definition:** A **measure space** is a triplet $(X, \mathscr{F}, \mu)$ where $\mathscr{F}$ is a $\sigma$-field of subsets of $X$ and $\mu$ is a measure defined on $\mathscr{F}$. A **measurable space** is a pair $(X, \mathscr{F})$.

## 2.1 Measurable functions

Let $(X, \mathscr{F})$ and $(Y, \mathscr{S})$ be two measurable spaces. The function $f : X \to Y$ is measurable if for all $B \in \mathscr{S}$, $f^{-1}(B) \in \mathscr{F}$. That is, the $\sigma$-algebra generated by $f$, $\sigma(f) = \{f^{-1}(B) \mid B \in \mathscr{S}\}$ is a subset of $\mathscr{F}$. Moreover, if $\mathscr{F} = \mathscr{B}(X)$ and $\mathscr{S} = \mathscr{B}(Y)$ are the Borel set of $X$ and $Y$, respectively, $f$ is called a **Borel measurable** function.

Let $\bar{\mathbb{R}}$ denote the set of the *extended real numbers*, $\mathbb{R} \cup \{\pm\infty\} = [-\infty, +\infty]$. We may define addition and multiplication as follows

1. $\forall a \in R, -\infty < a < \infty$.
2. $\forall a \in R, a + (\pm\infty) = \pm\infty$.
3. $\forall a \in \mathbb{R}^+, a(\pm\infty) = \pm\infty$.
4. $(-1)(\pm\infty) = \mp\infty$.

The extended Borels sets, $\mathscr{B}(\bar{\mathbb{R}})$ are collection of subsets having the following form

$$A, \quad A \cup \{\pm\infty\}, \quad A \cup \{-\infty, +\infty\}$$

where $A$ is a Borel set. The extended Borel set make a $\sigma$-field.

For the rest of this text, we may assume a measurable function $f : X \to \bar{\mathbb{R}}$ where $\bar{\mathbb{R}}$ is equipped with $\mathscr{B}(\bar{\mathbb{R}})$.

**Lemma 2.1.** *Suppose $f : X \to \bar{\mathbb{R}}$ is a function. The followings are equivalent*

1. *$f$ is measurable.*
2. *For all $a \in \mathbb{R}$, $\{x \in X \mid f(x) > a\} \in \mathscr{F}$.*
3. *For all $a \in \mathbb{R}$, $\{x \in X \mid f(x) \geq a\} \in \mathscr{F}$.*
4. *For all $a \in \mathbb{R}$, $\{x \in X \mid f(x) < a\} \in \mathscr{F}$.*
5. *For all $a \in \mathbb{R}$, $\{x \in X \mid f(x) \leq a\} \in \mathscr{F}$.*

**Example 2.1.** Let $f : \mathbb{R}^n \to \mathbb{R}$ and $\mathscr{F} = \mathscr{M}$ the Lebesgue measurable sets. If $f$ is continuous, then $f$ is measurable.

Random variables are measurable functions from a measure space.

**Theorem 2.2.** *If $f$ and $g$ are measurable functions, then* $\max(f, g)$ *and* $\min(f, g)$ *are also measurable.*

**Corollary 2.3.** *Suppose $f$ is a measurable function, then*

$$f^+(x) = \begin{cases} f(x) & f(x) \geq 0 \\ 0 & f(x) < 0 \end{cases} \qquad f^-(x) = \begin{cases} -f(x) & f(x) \leq 0 \\ 0 & f(x) > 0 \end{cases}$$

*are measurable function. Since $f = f^+ - f^-$, then every function is the difference of two non-negative measurable functions.*

**Definition:** Let $f_i$ be functions of $X$ to $\bar{\mathbb{R}}$. Then

$$\inf f_i(x) = \inf\{f_i(x)\} \qquad \sup f_i(x) = \sup\{f_i(x)\}$$

**Theorem 2.4.** *If $\{f_i\}$ are measurable functions, then* $\sup f_i$ *and* $\inf f_i$ *are measurable functions.*

Furthermore, we may define lim sup and lim inf as follows

$$\begin{aligned} \limsup f_i &= \lim_{n\to\infty} \sup_{i\geq n} f_i & \liminf f_i &= \lim_{n\to\infty} \inf_{i\geq n} \\ &= \inf_n \sup_{i\geq n} f_i & &\sup_n \inf_{i\geq n} f_i \end{aligned}$$

**Corollary 2.5.** *If $\{f_i\}$ is a collection of measurable functions, then* $\limsup f_i$ *and* $\liminf f_i$ *are measurable.*

**Corollary 2.6.** *Suppose $\{f_i\}$ are measurable and converge pointwise to $f$. Then, $f$ is measurable.*

**Theorem 2.7.** *Let $f_i : X \to \mathbb{R}$ be some measurable functions. Then, for a continuous function $G : \mathbb{R}^n \to \mathbb{R}$, $G(f_1, \ldots, f_n)$ is measurable.*

## 2.2 The Lebesgue Integral

The measurable function $s : X \to \mathbb{R}$ is a **step** function if it takes on only finite number of values.

**Example 2.2.** Let $E \in \mathscr{F}$ then

$$\chi_E(x) = \begin{cases} 1 & x \in E \\ 0 & x \notin E \end{cases}$$

is a step function. Furthermore, let $s$ be a simple function that takes on values $c_1, c_2, \ldots, c_n$ and let $E_i = s^{-1}(c_i)$ for $i = 1, \ldots, n$. Then,

$$s = \sum_{i=1}^{n} c_i \chi_{E_i}$$

**Definition:** Let $s : X \to \mathbb{R}$ be a (non-negative) simple function and let $c_1, \ldots, c_n$ be distinct non-zero values of $s$ with $E_i = s^{-1}(c_i)$. Let $E \in \mathscr{F}$ and define the *integral of $s$ over $E$ with respect to $\mu$* as the sum

$$\int_E s = \sum_{i=1}^{n} c_i \mu(E \cap E_i)$$

Note that, the integral might be $+\infty$ since $\mu(E \cap E_i)$ might be $+\infty$.

**Proposition 2.8.** *LEt $s$ and $r$ be simple non-negative functions and $E \in \mathscr{F}$.*

1. $\int_E s + r = \int_E s + \int_E r$.
2. $\int_E cs = c \int_E s$ *for* $c \geq 0$.
3. *If* $s \leq r$, *then* $\int_E s \leq \int_E r$.

**Definition:** Let $f : X \to \bar{\mathbb{R}}$ be a non-negative measurable function and $E \in \mathscr{F}$. The *integral of $f$ over $E$ with respect to $\mu$* is defined as

$$\int_E f \, \mathrm{d}\mu = \sup\left\{ \int_E s \,\middle|\, s \leq f, s \text{ is simple} \right\}$$

Let $s$ be a simple function. We need to check that the new definiton of integral is equivalent to the old one in the case of simple functions. That is

$$\int_E s \, \mathrm{d}\mu = \sum_{i=1}^{n} c_i \mu(E \cap E_i)$$

To justify why simple functions are used to approximate consider the following.

**Theorem 2.9.** *Let $f$ be a non-negative measurable function. Then, there exists a sequence of non-negative simple functions*

$$0 \leq s_1 \leq s_2 \leq \cdots \leq f$$

*such that $s_i \to f$ pointwise. Moreover, if $f$ is bounded, $s_i \rightrightarrows f$.*

*Proof.* Fix $n$ and divide the interval $[0, n[$ to $n2^n$ subinterval of length $2^{-n}$.

$$I_{n,i} = \left\{ \frac{i-1}{n2^n} \leq x < \frac{i}{n2^n} \right\} \qquad i = 1, \ldots n2^n$$

Then let $E_{n,i} = f^{-1}(I_{n,i})$ and $F_n = f^{-1}([n, +\infty])$. Note that $E_{n,i}$ and $F$ are mutually disjoint and cover $X$.

$$s_n(x) = \sum_{i=1}^{n2^n} \left( \frac{i-1}{2^n} \right) \chi_{E_{n,i}} + n\chi_{F_n}$$

then $s_n \leq f$ and $s_n \leq s_{n+1}$ for all $n$. ■

In contrast to Riemann integral we approximate by dividing the range of the function. Removing the conditions on $x$-axis. Gives good approximation without $f$ having to be continuous.

**Proposition 2.10.** *Let $f$ and $g$ be non-negative measurable functions and $E, F \in \mathscr{F}$. Then*

1. *If $f \leq g$, then $\int_E f \, \mathrm{d}\mu \leq \int_E g \, \mathrm{d}\mu$.*
2. *If $E \subset F$, then $\int_E f \, \mathrm{d}\mu \leq \int_F f \, \mathrm{d}\mu$.*
3. *If $\mu(E) = 0$, then $\int_E f \, \mathrm{d}\mu = 0$.*

**Theorem 2.11 (Chebyshev).** *Let $f$ be a non-negative measurable function and let $E \in \mathscr{F}$ and $c > 0$. Define $E_c = \{x \in E \mid f(x) \geq c\}$, then*

$$\mu(E_c) \leq \frac{1}{c} \int_E f \, \mathrm{d}\mu$$

**Corollary 2.12.** *Let $f$ be a non-negative measurable function with $\int_E f \, \mathrm{d}\mu < \infty$, then*

$$\mu(\{x \in E \mid f(x) = +\infty\}) = 0$$

**Definition:** If a property holds on a set $E \in \mathscr{F}$ except for a subset of zero measure, we say that the property holds **almost everywhere** on $E$.

**Corollary 2.13.** *Let $f$ be a non-negative function and $E \in \mathscr{F}$.*

$$\int_E f \, \mathrm{d}\mu = 0 \implies f \equiv 0 \text{ a.e. on } E$$

**Theorem 2.14.** *Let $f$ be a non-negative function and $A_1, A_2, \ldots$ pairwise disjoint from $\mathscr{F}$.*

$$\int_{\bigcup_{i=1}^{\infty} A_i} f \, \mathrm{d}\mu = \sum_{i=1}^{\infty} \int_{A_i} f \, \mathrm{d}\mu$$

We can use integrals to define measurse. *Gaussian measure* $\mu_G$ is defined on measurable subsets of $\mathbb{R}$

$$\mu_G(A) = \frac{1}{\sqrt{2\pi}} \int_A e^{-x^2/2} \, \mathrm{d}\mu_L$$

Moreover, $\mu_G$ is a probability measure.

**Corollary 2.15.** *Let $f$ and $g$ be a non-negative functions and $E \in \mathscr{F}$. If $f = g$ a.e.on $E$*

$$\int_E f \, \mathrm{d}\mu = \int_E g \, \mathrm{d}\mu$$

## 2.3 Further properties of integrals

Let $\{f_i\}$ be a sequence of measurable functions with

$$0 \leq f_1 \leq f_2 \leq \ldots$$

Then, $f = \lim_{n\to\infty} f_n$ exists and is measurable.

**Lemma 2.16.** *Let $f$ be a non-negative measurable function on $X$ and let $E_1, E_2, \ldots$ be a sequence of sets in $\mathscr{F}$ with $E_1 \subset E_2 \subset \ldots$ and $E = \cup_i E_i$. Then*

$$\int_E f \, \mathrm{d}\mu = \lim_{i\to\infty} f \, \mathrm{d}\mu$$

**Theorem 2.17 (Monotone convergence theorem).** *Let $f$ and $\{f_i\}$ be described as above. Then for $E \in \mathscr{F}$*

$$\int_E f \,\mathrm{d}\mu = \lim_{n\to\infty} f_n \,\mathrm{d}\mu$$

**Remark 4.** Let $f$ be a non-negative measurable function and $s_n$ be the step functions from the construction. By 2.17

$$\int_E s_n \,\mathrm{d}\mu \to \int_E f \,\mathrm{d}\mu$$

**Theorem 2.18.** *Suppose $f$ and $g$ are two non-negative measurable function, $c > 0$, and $E \in \mathscr{F}$*

1. $\int_E f + g \,\mathrm{d}\mu = \int_E f \,\mathrm{d}\mu + \int_E g \,\mathrm{d}\mu$.
2. $\int_E cf \,\mathrm{d}\mu = c\int_E f \,\mathrm{d}\mu$.

**Corollary 2.19.** *Let $\{f_i\}$ be non-negative measurable functions. Then, $\sum f_i$ is a non-negative measurable function and*

$$\int_E \sum_{i=1}^{\infty} f_n \,\mathrm{d}\mu = \sum_{i=1}^{n} \int_E f_n \,\mathrm{d}\mu$$

**Lemma 2.20.** *The following two conditions are equivalent*

1. $\int_E |f| \,\mathrm{d}\mu < +\infty$.
2. $\int_E f^+ \,\mathrm{d}\mu < +\infty$ *and* $\int_E f^- \,\mathrm{d}\mu < +\infty$.

*Proof.* $|f| = f^+ + f^-$. ■

**Definition:** A measurable function $f$ is **integrable** over $E$ if either of the conditions hold. In this case, $f \in \mathscr{L}(\mu, E)$. If $E = X$, then $f \in \mathscr{L}(\mu)$. For $f \in \mathscr{L}(\mu, E)$

$$\int_E f \,\mathrm{d}\mu = \int_E f^+ \,\mathrm{d}\mu - \int_E f^- \,\mathrm{d}\mu$$

**Theorem 2.21.** *Suppose $f, g \in \mathscr{L}(\mu, E)$ and $c \in \mathbb{R}$*

1. $cf \in \mathscr{L}(\mu, E)$ *and* $\int_E cd \,\mathrm{d}\mu = c\int_E f \,\mathrm{d}\mu$.
2. $f + g \in \mathscr{L}(\mu, E)$ *and* $\int_E f + g \,\mathrm{d}\mu = \int_E f \,\mathrm{d}\mu + \int_E g \,\mathrm{d}\mu$.
3. *If $f \le g$, then* $\int_E f \,\mathrm{d}\mu = \int_E g \,\mathrm{d}\mu$.

**Corollary 2.22.** *Let $f \in \mathscr{L}(\mu, E)$, then*

$$\left|\int_E f \,\mathrm{d}\mu\right| \le \int_E |f| \,\mathrm{d}\mu$$

**Lemma 2.23 (Fatou's lemma).** *Assume $f_1, f_2, \ldots$ are non-negative measurable function and let $f = \liminf f_n$*

$$\int_E f \,\mathrm{d}\mu \le \liminf \int_E f_n \,\mathrm{d}\mu$$

**Theorem 2.24 (Lebesgue dominated converge theorem).** *Let $f_1, f_2, \ldots$ be a sequence of measurable functions and let $E \in \mathscr{F}$. Suppose the following assumptions hold*

1. $\lim_{n\to\infty} f_n(x)$ *exists for all* $x \in E$.
2. *There is a non-negative measurabe function* $g \in \mathscr{L}(\mu, E)$ *with* $g \geq |f_n|$ *on* $E$ *for all* $n$.

*Then,* $f(x) = \lim_{n\to\infty} f_n(x)$ *is integrable and*

$$\int_E \lim_{n\to\infty} f_n \,\mathrm{d}\mu = \lim_{n\to\infty} \int_E f_n \,\mathrm{d}\mu$$

**Corollary 2.25.** *Let $\{f_i\}$ be a sequence of functions in $\mathscr{L}(\mu, E)$ with*

$$\sum_{i=1}^{\infty} \int_E |f_n| \,\mathrm{d}\mu < +\infty$$

*Then,*

1. $\sum f_n$ *converges absolutely a.e.on* $E$ *and is integrable on* $E$.
2. $$\int_E \sum_{n=1}^{\infty} f_n \,\mathrm{d}\mu = \sum_{n=1}^{\infty} \int_E f_n \,\mathrm{d}\mu$$

## 2.4 Lebesgue integral vs Riemann integral

**Theorem 2.26.** *Let $f$ be a bounded Riemann integrable function on $[a, b]$ with Riemann integral $\int_a^b f(x)\,\mathrm{d}x$. Then, $f \in \mathscr{L}(\mu_L, [a, b])$ and*

$$\int_a^b f \,\mathrm{d}x = \int_{[a,b]} f \,\mathrm{d}\mu_L$$

## 2.5 Fubini theorem

Let $(X, \mathscr{M}, \mu)$ and $(Y, \mathscr{N}, \nu)$ be two measure spaces. Let $X \times Y$ demote the space

$$X \times Y = \{(x, y) \,|\, x \in X, y \in Y\}$$

**Definition:** $A \times B \subset X \times Y$ is a **product set** if $A \in \mathscr{M}$ and $B \in \mathscr{M}$. The smallest $\sigma$-field in $X \times Y$ containing all product sets $A \times B$ is denoted by $\mathscr{M} \otimes \mathscr{N}$.

**Definition:** For $E \subset X \times Y$ and fix $x \in X$, let $E_x = \{y \in Y \,|\, (x, y) \in E\}$. $E_x$ is called the $x$-slice of $E$.

**Proposition 2.27.** *IF $E \in \mathscr{M} \otimes \mathscr{N}$, then $E_x \in \mathscr{N}$.*

**Corollary 2.28.** *LEt $f : X \times Y \to \bar{\mathbb{R}}$ be meaurable with respect to $\mathscr{M} \otimes \mathscr{N}$. For fixed $x_0 \in X$, define $f_{x_0} : Y \to \mathbb{R}$ given by $f_{x_0}(y) = f(x_0, y)$. Then, for eah $x_0 \in X$, $f_{x_0}$ is a measurable function on $Y$.*

Suppose $X$ and $Y$ are $\sigma$-finite. We now make a measure on $\mathscr{M} \otimes \mathscr{N}$ using $\mu$ and $\nu$.

**Definition:** Let $Z$ ba set and let $\mathscr{S}$ be a collection of subsets of $Z$. $\mathscr{S}$ is called a $\lambda$-system if the following three properties hold.

$\lambda 1$. $Z \in \mathscr{S}$.

$\lambda 2$. If $E_1 \subset E_2 \subset \ldots$ is an increasing sequence with each $E_n \in \mathscr{S}$, then

$$\bigcup_{i=1}^{\infty} E_n \in \mathscr{S}$$

$\lambda 3$. If $E, F \in \mathscr{S}$ and $E \subset F$, then $F - E \in \mathscr{S}$.

**Definition:** Let $\mathscr{P}$ be a collection of subsets of $Z$. $\mathscr{P}$ is called a $\pi$-system if the following property holds.

$\pi 1$. If $A, B \in \mathscr{P}$, then $A \cap B \in \mathscr{P}$.

**Theorem 2.29 (Dynkin $\boldsymbol{\pi}$-$\boldsymbol{\lambda}$ theorem).** *If $\mathscr{S}$ is a $\lambda$-system and $\mathscr{P}$ is a $\pi$-system with $\mathscr{P} \subset \mathscr{S}$, then the smallest $\sigma$-field containing $\mathscr{P}$, $\sigma(\mathscr{P})$ is contained in $\mathscr{S}$.*

**Proposition 2.30.** *If $E \in \mathscr{M} \otimes \mathscr{N}$ and $\phi_E : X \to \mathbb{R}$ is defined by $\phi_E(x) = \nu(E_x)$, then $\phi_E$ is measurable.*

**Definition:** Let $E \in \mathscr{M} \otimes \mathscr{N}$ and define

$$\pi'(E) = \int_X \phi_E(x)\, \mathrm{d}\mu$$

to be the product measure on $E$.

**Proposition 2.31.** *$\pi'$ is a measure.*

Suppose instead of $x$-slices we used $y$-slices and denoted the measure by $\pi''$.

**Theorem 2.32 (Fubini, version 1).**

$$\pi' = \pi''$$

**Definition:** The measure $\pi' = \pi''$ is denoted by $\mu \times \nu$ and is called the product measure on $\mathscr{M} \otimes \mathscr{N}$.

**Example 2.3.** Let $X = Y = \mathbb{R}$ and $\mathscr{M} = \mathscr{N} = \mathscr{B}(\mathbb{R})$. Also, let $\mu = \nu = \mu_L$ the Lebesgue measure on $\mathbb{R}$. We claim that, $\mathscr{M} \otimes \mathscr{N} = \mathscr{B}(\mathbb{R}^2)$ and $\mu \times \nu = \mu_L^2$, the Borel sets and Lebesgue measure in $\mathbb{R}^2$.

**Example 2.4.** We can show that

$$\mathscr{B}(\mathbb{R}^n) \otimes \mathscr{B}(\mathbb{R}^m) = \mathscr{B}\big(\mathbb{R}^{m+n}\big)$$

and $\mu_L^m \times \mu_L^n = \mu_L^{m+n}$.

**Theorem 2.33 (Fubini, version 2).** *Let $f : X \times Y \to \mathbb{R}$ be a non-negative measurable function. Then*

1. *For each $x_0 \in X$, $f(x_0, y)$ is a measurable function of $y$.*
2. *For each $y_0 \in X$, $f(x, y_0)$ is a measurable function of $x$.*
3. *$\int_Y f(x, y)\,\mathrm{d}\nu$ is a measurable function of $x$.*
4. *$\int_X f(x, y)\,\mathrm{d}\mu$ is a measurable function of $y$.*
5. $$\int_{X\times Y} f(x, y)\,\mathrm{d}\mu \times \nu = \int_X \int_Y f(x, y)\,\mathrm{d}\nu\,\mathrm{d}\mu = \int_Y \int_X f(x, y)\,\mathrm{d}\mu\,\mathrm{d}\nu$$

**Theorem 2.34 (Fubini, version 2).** *Let $f : X \times Y \to \mathbb{R}$ be an integrable function. Then*

1. *For almost all $x \in X$, $f(x, y)$ is a integrable function of $y$.*
2. *For almost all $y \in X$, $f(x, y)$ is a integrable function of $x$.*
3. *$\int_Y f(x, y)\,\mathrm{d}\nu$ is equal a.e.to an integrable function on $X$.*
4. *$\int_X f(x, y)\,\mathrm{d}\mu$ is equal a.e.to an integrable function on $Y$.*
5. $$\int_{X\times Y} f(x, y)\,\mathrm{d}\mu \times \nu = \int_X \int_Y f(x, y)\,\mathrm{d}\nu\,\mathrm{d}\mu = \int_Y \int_X f(x, y)\,\mathrm{d}\mu\,\mathrm{d}\nu$$

## 2.6 Random variables, expectation values, and indepedence